Dhruv Milind Thakur

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.58
Seth Hirachand Mutha college kalyan(west)	Class XII (HSC Board Examinations in Maharashtra State)	2024	76.00%
Seth Hirachand Mutha School kalyan(west)	Class X (CBSE)	2022	83.20%

OPEN SOURCE CONTRIBUTION

Contributed to the Supabase Auth Go backend by implementing exact, case-insensitive user lookup by email on the admin API; bypassed slow fuzzy searches with index-supported PostgreSQL queries and verified functionality using Go unit testing and Docker environments to support client SDKs. | PR #2668

Contributed to Rakuten Tech's QueryCraft AI-SQL assistant by implementing offline Prisma ORM schema parsing; designed a TypeScript transpiler to convert Prisma models into SQL DDL for the LLM context, bypassed Next.js API credential validations, and verified compilation using TypeScript. | PR #109

PROJECTS

High-Throughput Wallet & Ledger System | Node.js (Fastify), PostgreSQL, Redis, Apache Kafka, Docker | GitHub

- **Reduced core transaction API latency by 98.10% (1526ms to 29ms)**, by offloading notification and logging tasks to asynchronous **Apache Kafka consumer groups**.
- **Eliminated write deadlocks and balance drift under a load of 166+ TPS** (throughput locally), by engineering a PostgreSQL **double-entry ledger** with alphabetical **lexicographical row-locking** (SELECT FOR UPDATE).
- **Prevented double-spending and duplicate requests in under 1ms**, by implementing a fast-path idempotency gate using **Redis distributed locks** (NX/PX keys) and write-through balance caches.

AI Financial Analyst | Python (FastAPI), Qdrant, BM25, PyTorch, QLoRA, Ollama | GitHub

- **Evaluated an LLM financial analyst across 35 benchmark cases**, achieving **100%** execution success and tool-selection accuracy through schema-aware SQL generation, connection-level validation filters, and context controls.
- **Designed a hybrid RAG pipeline** (achieving **83.33%** recall and **88.57%** citations) and **fine-tuned a local 1B model via QLoRA**, boosting its task success rate from **40% to 91.43%** (+51.43 pp) and reducing latency by **17%.**

Financial Intelligence Platform | Node.js, Python (Scikit-Learn, Pandas) | GitHub | Link

- **Engineered a schema-compliant auditing and segmentation pipeline**, verifying **1,800+ transactions** for **100% balance correctness** via sequential double-entry replays and using **K-Means** to cluster users into RFM cohorts.
- Built a customizable **ML engine** and React dashboard using **scikit-learn** Isolation Forest (isolating **3% outliers**) and Recursive Random Forest (**94% R^2 accuracy**) to project 30-day platform cashflows.

MapLink | React.js, Node.js (Express), Socket.io, PostgreSQL, PostGIS, Redis, Nginx, Docker, GitHub Actions | GitHub

- Reduced database write transactions by **98%** and query latency by **97%** for historical routes by implementing a **Redis Write-Behind Cache** (15s bulk flusher) combined with PostgreSQL composite and **GiST spatial indexes**
- Reduced client location requests by **75%** using a **Haversine filter** with a **3s update interval** and **5m threshold**
- Designed a **horizontally scaled backend cluster** with 3 Node.js instances load-balanced by **Edge Nginx (sticky ip_hash)**, integrating a **Redis Pub/Sub Adapter** to sync WebSocket broadcasts with an **8ms cross-server latency**.

SKILLS

Programming: C/C++, JavaScript, Python

Frameworks & Libraries: Node.js, Fastify, Express.js, React.js, Pandas, NumPy, Scikit-learn, PyTorch

AI/LLM: LLMs, RAG, Embeddings, Qdrant, BM25, QLoRA, Ollama

Databases & Backend: PostgreSQL, MongoDB, Redis, REST APIs, WebSockets, JWT, Kafka

Tools & DevOps: Docker, Nginx, Git, GitHub Actions, AWS S3, Postman

POSITIONS OF RESPONSIBILITY

Head, Institute Technical Council - Operations Domain

Assessment Lead, Office of Career Council, IIT Hyderabad

Core, Management and Operation, Tinkerers' lab, IIT Hyderabad

- Managed the Operational execution for procurement of **Tinkerers**' lab
- Directed technical programs, conducted/executed events, and managed administration for the Thrust tech event.

RELEVANT COURSES

Computer Science & Mathematics: Introduction to Programming, Data Structures and Algorithms, Introduction to AI, Linear Algebra, Introduction to Probability, Introduction to Statistics, Basic Calculus

Physics & Engineering: Computational Physics, Quantum Mechanics, Analog Electronics, Digital Circuits.

EXTRACURRICULAR

Leetcode - Solved **500+ Data Structures and Algorithms** problems on LeetCode

My Portfolio - Link | GitHub

Sports - Playing football under NSO IIT Hyderabad

EXTRA COURSES

Introduction to Generative AI | Google Cloud | Certificate

Get Started with Databricks for Generative AI | Databricks | Certificate